Pranav M

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Education

2023 – 2027 Electronics and Communication Engineering at **PES University**
2022 Class 12th at **Vydehi School of Excellence**

Projects

Reconfigurable DDR5 Memory Controller (RMC) GitHub
Architected a parameterized DDR5 memory controller targeting JEDEC JESD79-5 and DFI 5.2, with an AXI4 host interface and scalable multi-channel/multi-rank support. Defined microarchitecture across the client interface, transaction tracking and ordering, MC-core datapath, APB/CSR interface, and DDR5 mode-register and power-management flows; contributed read-path verification and a DFT observation interface.

HADES – Programmable Processing-Near-Memory Engine GitHub
Designed and verified a programmable processing-near-memory engine for binary similarity search, with a 12-opcode, 16-bit ISA and dual-row/selective-scan mechanisms around a shared 2R1W register-file macro. Submitted for tapeout through the Tiny Tapeout SKY130 flow; post-layout timing closes at 40 MHz with zero reported DRC, LVS, and antenna violations.

Specula – Out-of-Order RISC-V Core GitHub
Implemented a single-issue out-of-order RISC-V core with physical-register renaming, a 32-entry reorder buffer, an 8-entry reservation station, and gshare/BTB branch prediction with misprediction recovery. Developed a store queue with byte-lane store-to-load forwarding and integrated Sv32 virtual memory and simulated UART, PLIC, and VirtIO block-device support. Five standalone benches and 16 directed programs passed in Bluesim; xv6-rv32 reached an interactive shell in simulation. Vivado synthesis reported +0.996 ns WNS at a 90 MHz constraint.

Aletheia – Processing-Near-Memory Framework GitHub
Built a PNM framework on a Raspberry Pi 5, dispatching commands over TCP to a memory node to execute computation locally. Ran PMU-backed characterization of working sets, stride patterns, and dataset scaling to study memory locality and cache behavior.

Herald – Fixed-Point DSP Coprocessor GitHub
Designed and taped out a Q12.12 DSP coprocessor through Tiny Tapeout, featuring CORDIC and multiply-accumulate (MAC) engines for FPU-less microcontrollers. Verified using Bluespec and cocotb.

Experience

PES University – Teaching Assistant, Computer Organization & Design Aug 2026 – Present
Teach RISC-V assembly and processor microarchitecture, and guide students through the design, implementation, simulation, and debugging of single-cycle RISC-V processors.

HSP, PES University – Mentor Jul 2025 – Present
Mentored students on digital hardware design projects across two consecutive terms, guiding architecture design, reviewing HDL, and debugging design modules.

Skills

Languages	SystemVerilog, Verilog, Bluespec, Veryl, C, Python, Assembly
Tools & Platforms	Vivado, GTKWave, Verilator, Icarus Verilog, cocotb, ModelSim, Git, Linux
Domains	Computer Architecture, Memory Systems, DDR/Memory Interfaces, Processing-Near-Memory, RTL Design and Verification, SoC Design